

The Universal Settlement Layer

The next layer of the internet — focused on data sovereignty and societal flourishing.

Budlum does not replace existing networks: it is a Layer-1 protocol that establishes a single agreed fact above chains, data and verification. This litepaper explains what the protocol does; its subject is the protocol's own features — not the applications that may be built on it.

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1. THE PROBLEM

Today's layers of the internet carry data and execute computation — but none of them records who owns which byte, whether a computation was performed correctly, or which participants agree on the same fact. The result is shifting ground beneath individuals and institutions alike:

Lost ownership

Identity, content, reach and revenue pooled inside the vaults of large platforms. The account belongs to the platform; the data belongs to the platform. Leaving means leaving everything behind.

A fragile information space

Who produced a media artifact, and when, cannot be verified; impersonation and unattributed reuse are routine. The AI outputs shaping decisions are unauditable black boxes.

Fragmented infrastructure

More than 20,000 chains cannot speak to each other. Institutional networks trade trust for speed; open networks trade speed for trust — and none can meet on a common final record.

The quantum time bomb

Deployed signatures (ECDSA, Ed25519) fall within NIST's 2030–2035 horizon. "Harvest now, decrypt later" is already underway, and migration day historically arrives as a crisis hard fork.

2. THE ANSWER — THE UNIVERSAL SETTLEMENT LAYER

Budlum adds one more layer beneath this ground: the **Universal Settlement Layer**. Every domain keeps its own consensus; Budlum verifies each domain's finality proof and settles the result onto a single record — the `GlobalBlockHeader`. It is TCP/IP for value: it does not ask how a network works inside; it records that the packet is true.

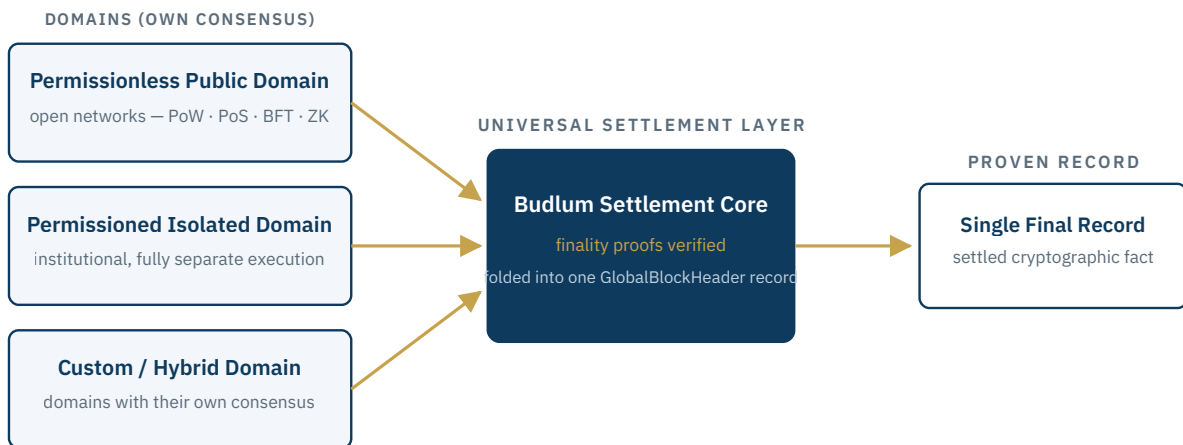


Figure 1 — Domains keep their own consensus; Budlum settles finality proofs into one final record.

The positioning is explicit: **the core is a single, monolithic settlement ledger**; modularity lives at the adapter boundary where domains attach. "Which chain are you on?" stops mattering.

3. KEY FEATURES

Each item below is a feature of the protocol itself: it would exist on Budlum even if no application were ever built.

Multi-consensus architecture

Every domain keeps its own consensus — proof-of-work, proof-of-stake, proof-of-authority, BFT or zero-knowledge. The layer settles all of them into one final record; cross-domain value moves through a proof-gated bridge lifecycle (lock → mint → burn → unlock).

Isolated institutional domain

A permissioned execution environment for regulated institutions, **fully separate** from the public network: the institution defines membership and rules, while the settlement fact is still anchored to the same layer.

Stake-based verifier registry

Verifiers earn roles by posting stake collateral and thereby render any content or transaction accountable. Misbehavior carries a protocol-enforced economic cost; collateral is slashed when a proof deadline is missed.

Data sovereignty

Assets are portable; data remains in the network's decentralized storage layer (B.U.D.) — no platform can lock you in. Access scopes (single-read, time-boxed, perpetual), single-use consumption records and revocation are on-chain protocol objects; keys are delivered with standard encryption (HPKE).

ZK-backed verifiable computation

An in-tree zero-knowledge virtual machine (BudZKVM) offers provable execution callable from contracts: the computation's trace is its proof — no third party needs to be trusted.

Accountable AI inference

On-chain model registrations and threshold-agreement attestations: which verifier majority attested an output is on record, and contracts can request inference directly. A verifier in the minority forfeits the reward share — never the stake.

4. VISION

Budlum unites two commitments in one architecture: **data sovereignty** — every byte's owner and access right known to the protocol; and **societal flourishing** — a commons where identity is portable, content has an owner, and verification is accountable. The internet's next layer is only meaningful when both are built together.

These features are inherited by any application that may be built on the protocol, including social media; applications are not the subject of this document. Token economics and the roadmap are published as separate documents.

Budlum is a v0.3-dev controlled devnet candidate; it is not audited mainnet software. Nothing herein is investment advice.

Budlum — we settle what others compute.

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